

CERTIFICATE OF CALIBRATION

ISSUED BY: Industrial Measurement & Calibration (Pvt) Ltd

CERTIFICATE NUMBER:

M-2025-5876

DATE OF ISSUE:

2025-10-24



CL-148



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ISO/IEC 17025:2017 Accredited Laboratory

DATE OF CALIBRATION:

2025-10-24

CUSTOMER DESCRIPTION

M.A.Pathiraja,
Sri Dharmadara Road, Hegoda ,
Rathgama , Galle ,

DESCRIPTION OF UNIT UNDER CALIBRATION (UUC)

Item:	Temperature & Humidity Meter
Make/ Model:	--- / ---
Serial number/Identification number:	--- / ---
Instrument range:	(0 to +50) °C
Division/Graduation:	0.01 °C
Resolution	0.01 °C
Location of calibration:	IMC Lab

TRACEABILITY INFORMATION

This calibration establishes metrological traceability to the SI unit of temperature (kelvin) defined by the International Temperature Scale of 1990 (ITS-90). The laboratory standard(s) used in this calibration – which are listed below- are regularly calibrated by an accredited calibration laboratory, ensuring unbroken traceability to the SI. To maintain reliability, these standards are monitored between calibrations using measurement assurance techniques, including check standards, control charts, and ice point verifications. This calibration complies with ISO/IEC 17025, ensuring valid and traceable measurement results.

Type	Reference no.	Certificate no.	Expiry date
Digital thermometer with Pt1000 sensor	LAB/TM-21(CH1)	(7424)362-0191A	2028-01-03

CALIBRATION METHOD

The test thermometer (UUC) was evaluated against a laboratory standard platinum resistance thermometer utilizing a temperature bath. The bath temperature was consistently stabilized at each designated point to facilitate a comparison between the test thermometer and the reference thermometer measurements. Comprehensive calibration was conducted in accordance with the laboratory-developed test method TM-02 - Thermometer calibration.

THE EXPANDED UNCERTAINTY, U , ASSOCIATED WITH THE ESTIMATE OF THE CORRECTED VALUE

The reported expanded uncertainty has been determined in accordance with requirements of the ISO Guide to the Expression of Uncertainty in Measurement (the 'GUM'). The expanded uncertainty is calculated by combining all relevant standard uncertainties and multiplying the combined standard uncertainty by a coverage factor $k = 2$, which corresponds to an approximate confidence level of 95%. However, in cases where the calculated expanded uncertainty falls below the Calibration and Measurement Capability (CMC) value stated in the laboratory's accreditation scope, the reported expanded uncertainty has been limited to the CMC value.

CALIBRATION RELATED DATA

Range calibrated: (+10 to +50) °C
 Temperature during calibration: (22 ± 1) °C
 Calibration due date (suggested by the customer): 2026-10-23

CALIBRATION RESULTS ⁽¹⁾Temperature measurement ⁽¹⁾

Reference Temperature °C	UUC Reading ⁽²⁾ °C	Correction °C	Expanded uncertainty ⁽³⁾ °C
9.79	10.33	-0.54	0.20
19.60	20.06	-0.46	0.20
29.67	29.91	-0.24	0.20
39.90	40.35	-0.45	0.20
49.89	50.28	-0.39	0.20

Humidity Measurement*

Reference Humidity (%)	UUC Reading(%)	Correction(%)	Expanded uncertainty (%)
55.00	53.21	1.79	±3
72.00	73.15	-1.15	±3

¹ All reported values were rounded-off to 0.01.

² Average of six repeated measurements of the thermometer (UUC).

³ The measurement uncertainty was evaluated taking into account the contributions from the following sources:

- Standard uncertainty of the calibration standard,
- Standard error of the regression in the determination of uncertainty of the standard,
- Estimated drift of the calibration standard,
- Resolution of the calibration standard and the UUC,
- Uncertainty in mean correction,
- Spatial variation of the temperature bath,
- Stability of the temperature bath,
- Rounding-off results.

* Not Covered under the scope of Accreditation

NOTES:

- The results specified on this certificate pertain solely to the item assessed.
- The calibration results reported in the certificate are valid at the time of and under the conditions of measurement.
- This report may not be reproduced except in full without the express written permission of the issuing authority.

Calibration performed by:

Sudesh Thanusara - Assistant technical Officer



APPROVED SIGNATURE

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